Source-Aware Dynamic Pricing with Strategic Buyers

Jinjiang Zou

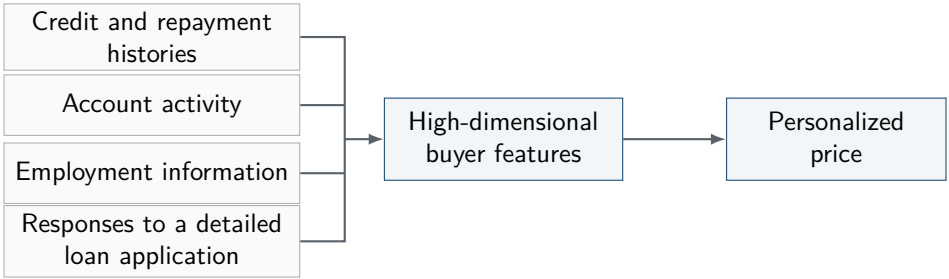
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High-dimensional features in personalized pricing

Effective personalized pricing depends on accurately learning how buyer features affect willingness to pay.

Example: consumer lending



A borrower's profile can include hundreds or thousands of candidate features.

Buyer features come from verified sources and buyers' own reports.

Javanmard and Nazerzadeh (2019); Wang et al. (2024); Liu et al. (2025).

Feature sources and strategic manipulation

These differences in feature sources affect buyers' opportunities for strategic manipulation.

Example: automobile insurance

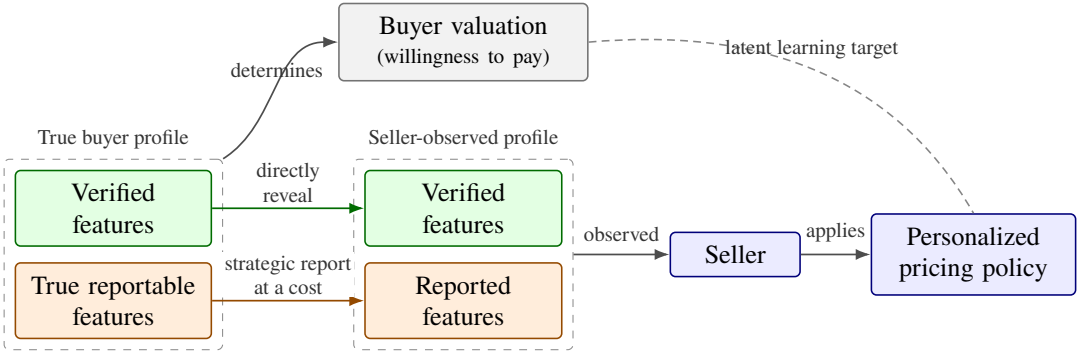
	Reportable features	**Verified features**
Buyer information	Self-reported odometer mileage	Verified claims histories
Strategic manipulation	Underreport mileage to seek a lower premium	Cannot be altered through misreporting

Strategic reporting makes it harder to learn how true buyer features affect willingness to pay.

The seller must protect valuation learning while making full use of verified information.

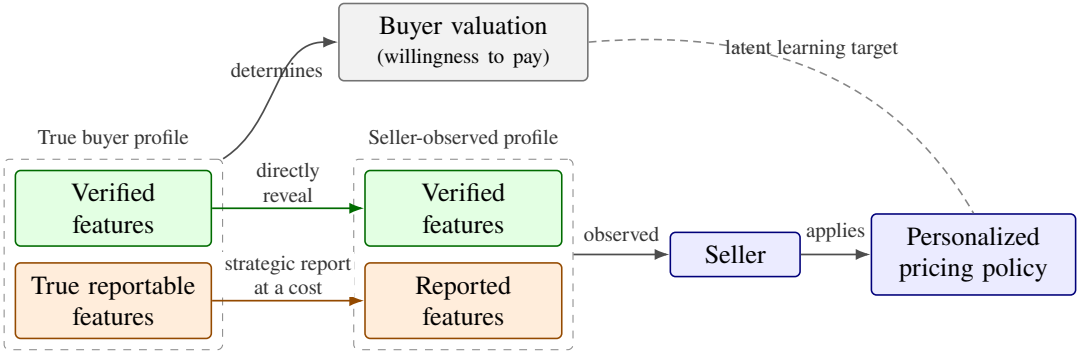
Gai and Puntoni (2021); Jeziorski et al. (2019); Liu et al. (2025).

Information structure in source-aware personalized pricing



- Buyer valuation depends on the **true features**.
- The seller observes verified features and potentially manipulated reports.

Information structure in source-aware personalized pricing



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Liu, Yang, Wang, and Sun (2025)
Contextual Dynamic Pricing with Strategic Buyers

- Low-dimensional buyer features and a known market-noise distribution.
- Feature-independent exploration collects truthful observations.
- Exploitation prices correct for strategic manipulation.

These policies do not explicitly distinguish verified from reportable features.

Verified information can support more refined personalized exploration.

Fan, Guo, and Yu (2024)
Policy Optimization Using Semiparametric Models for Dynamic Pricing

- Linear valuations, an unknown market-noise distribution, and binary purchase feedback.
- Randomized exploration supports valuation learning and nonparametric noise estimation.
- The analysis includes a high-dimensional extension.

This work does not account for strategic feature reporting.

We learn valuations and the pricing function while accounting for strategic reporting.

The high-dimensional extension uses penalized variable selection followed by least-squares refitting.

Main Contributions

A source-aware pricing framework. We distinguish verified from reportable features in a high-dimensional semiparametric setting with strategic buyers.

Source-Aware Localized Exploration and Calibrated Exploitation (SALE). Verified features localize exploration to buyer-specific pricing intervals. Both verified and reported features enter calibrated exploitation prices.

Learning and regret guarantees. We establish sparse and nonparametric estimation guarantees under non-i.i.d. buyer arrivals, and sublinear expected regret with an explicit gain from localization.

At time t, a buyer arrives with two types of true features:

$$\mathbf{x}^0_{t,\mathrm{V}} \in \mathcal{B}_{\mathrm{V}} \subseteq \mathbb{R}^{d_{\mathrm{V}}}, \qquad \mathbf{x}^0_{t,\mathrm{R}} \in \mathcal{B}_{\mathrm{R}} \subseteq \mathbb{R}^{d_{\mathrm{R}}}.$$

- The seller observes the verified features directly.
- The buyer may strategically manipulate the reportable features from $\mathbf{x}^0_{t,\mathrm{R}}$ to $\mathbf{x}_{t,\mathrm{R}}$ to seek a lower price.

After receiving the reported features, the seller observes

$$\mathbf{x}_t = \left((\mathbf{x}^0_{t,\mathrm{V}})^\top, \mathbf{x}_{t,\mathrm{R}}^\top, 1\right)^\top.$$

The reporting decision and manipulation cost involve only reportable features.

Let $d = d_{\mathrm{V}} + d_{\mathrm{R}}$ and denote the buyer's full true feature vector by

$$\mathbf{x}_t^0 = \left((\mathbf{x}_{t,\mathrm{V}}^0)^\top, (\mathbf{x}_{t,\mathrm{R}}^0)^\top, 1\right)^\top.$$

The buyer's valuation follows a sparse high-dimensional linear model:

$$v_t = \boldsymbol{\theta}_0^\top \mathbf{x}_t^0 + z_t, \qquad \boldsymbol{\theta}_0 = (\boldsymbol{\beta}_{\mathrm{V},0}^\top, \boldsymbol{\beta}_{\mathrm{R},0}^\top, \alpha_0)^\top, \qquad s_0 = \|\boldsymbol{\theta}_0\|_0 \ll d.$$

- Market noise z_t is mean-zero and independent of the features. The noises are i.i.d. with unknown cdf F supported on $(-\delta_z, \delta_z)$.
- Buyer valuations satisfy $v_t \in [\delta_v, B - \delta_v]$. The seller posts $p_t \in [0, B]$ and observes $y_t = \mathbb{I}\{p_t \le v_t\}$.
- Buyer features may be temporally dependent under predictable context stability.

The horizon T may be unknown to the seller.

Let $u_t = \boldsymbol{\theta}_0^\top \mathbf{x}_t^0$ denote the buyer's valuation excluding market noise. The corresponding expected revenue is

$$\mathrm{rev}_t(p) = p\{1 - F(p - u_t)\}.$$

The clairvoyant seller knows $(\boldsymbol{\theta}_0, F, \mathbf{x}_t^0)$ and posts

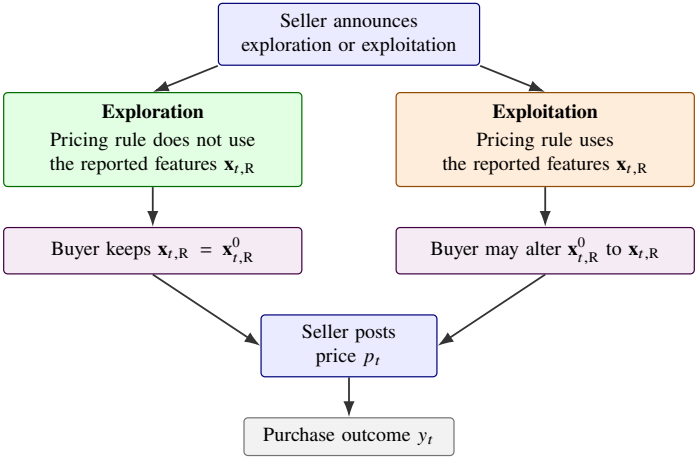
$$p_t^* = g(u_t), \qquad g(u) \in \arg\max_{p \in [0,B]} p\{1 - F(p - u)\}.$$

We measure performance by expected cumulative regret relative to the oracle:

$$\boxed{\mathrm{Regret}_\pi(T) = \mathbb{E}_\pi \sum_{t=1}^{T} \{\, \mathrm{rev}_t(p_t^*) - \mathrm{rev}_t(p_t)\}.}$$

If the maximum expected revenue is zero, set $g(u) = 0$.

Pricing Rules in Exploration and Exploitation



Before the buyer reports, the seller announces the pricing phase and discloses the form of the corresponding pricing rule.

Similar to Liu et al. (2025), the buyer knows $\boldsymbol{\theta}_0$ and F, and hence g, whereas $\boldsymbol{\theta}_0$ and F are unknown to the seller. The seller discloses the pricing rule but not the estimates used in exploitation.

Without access to these estimates, the buyer uses her known valuation information to form the anticipated price. Her optimal report balances this price against the manipulation cost:

$$\mathbf{x}_{t,\mathrm{R}} \in \underset{\mathbf{x}_{\mathrm{R}} \in \mathbb{R}^{d_{\mathrm{R}}}}{\arg\min} \left\{ \underbrace{g\big(\boldsymbol{\beta}_{\mathrm{V},0}^{\top}\mathbf{x}_{t,\mathrm{V}}^{0} + \boldsymbol{\beta}_{\mathrm{R},0}^{\top}\mathbf{x}_{\mathrm{R}} + \alpha_0\big)}_{\text{anticipated price}} + \underbrace{\frac{1}{2}(\mathbf{x}_{\mathrm{R}} - \mathbf{x}_{t,\mathrm{R}}^{0})^{\top}\mathbf{A}(\mathbf{x}_{\mathrm{R}} - \mathbf{x}_{t,\mathrm{R}}^{0})}_{\text{manipulation cost}} \right\}. \tag{1}$$

Here $\mathbf{A} \in \mathbb{R}^{d_{\mathrm{R}} \times d_{\mathrm{R}}}$ is symmetric positive definite and common across buyers.

Only reportable features are chosen; verified features remain fixed.

Under the interior-response conditions, the first-order condition of (1) gives

$$\mathbf{x}_{t,\mathrm{R}} = \mathbf{x}^0_{t,\mathrm{R}} - \mathbf{A}^{-1}\boldsymbol{\beta}_{\mathrm{R},0}\, g'(\boldsymbol{\theta}_0^\top \mathbf{x}_t). \tag{2}$$

This relates the true and reported valuation components:

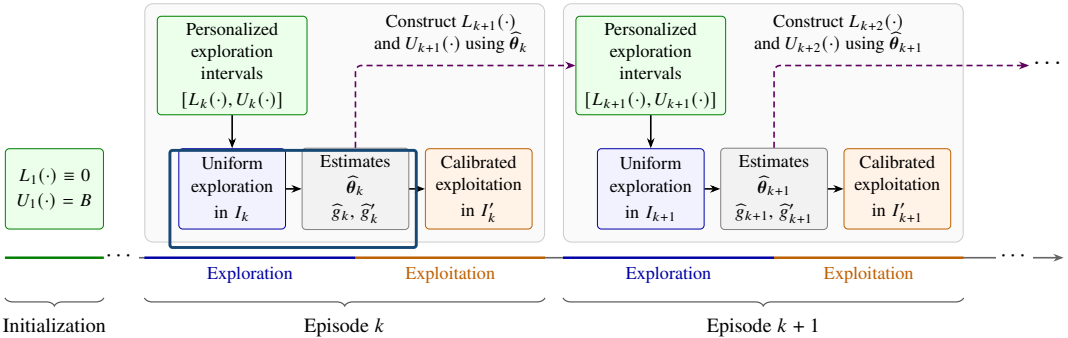
$$\boldsymbol{\theta}_0^\top \mathbf{x}_t^0 = \underbrace{\boldsymbol{\theta}_0^\top \mathbf{x}_t}_{\text{reported features}} + \underbrace{\boldsymbol{\beta}_{\mathrm{R},0}^\top \mathbf{A}^{-1}\boldsymbol{\beta}_{\mathrm{R},0}\, g'(\boldsymbol{\theta}_0^\top \mathbf{x}_t)}_{\text{strategic correction}}. \tag{3}$$

The seller corrects the valuation computed from reported features before applying the pricing function g.

Implementing this correction requires estimates of θ_0, g, and g'.

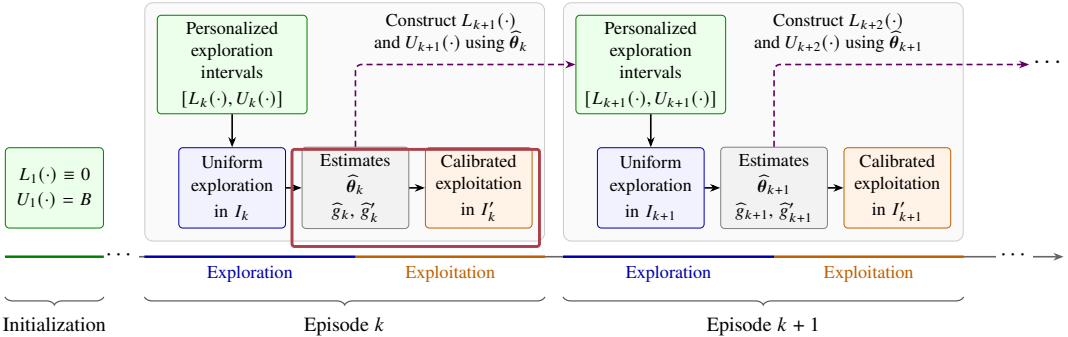
The response stays in the regular pricing region under Assumption 5.

SALE-K operates in episodes of increasing length, $\ell_k = 2^{k-1}\ell_0$. Each episode consists of exploration I_k, followed by exploitation I'_k.



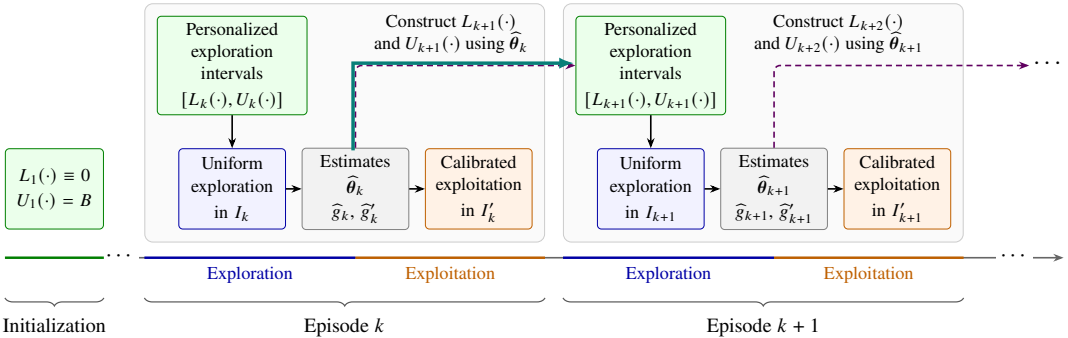
Exploration. Draw prices from the personalized intervals and use purchase feedback to estimate $\boldsymbol{\theta}_0$, g, and g'.

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Current exploitation. Use these estimates to calibrate prices for strategic manipulation in the same episode.

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Next exploration. Use $\widehat{\boldsymbol{\theta}}_k$ to construct L_{k+1}, U_{k+1} before episode $k+1$ begins.

Algorithm 1 SALE-K (Known $\mathbf{A}$): exploration

Input: B, initial episode length ℓ_0, and $\mathbf{A}$.
Initialize $L_1 \equiv 0$ and $U_1 \equiv B$.
For each episode $k = 1, 2, \ldots$**:**
Set $\ell_k = 2^{k-1}\ell_0$ and $S_k = 1 + \sum_{j<k} \ell_j$.
Set the exploration duration a_k according to (8).
Set $I_k = \{S_k, \ldots, S_k + a_k - 1\}$ and $I'_k = \{S_k + a_k, \ldots, S_k + \ell_k - 1\}$.
for $t \in I_k$ **do**
    Observe $\mathbf{x}^0_{t,\mathrm{V}}$ and the truthful report $\mathbf{x}_{t,\mathrm{R}} = \mathbf{x}^0_{t,\mathrm{R}}$.
    Draw $p_t \sim \mathrm{Unif}(L_k(\mathbf{x}^0_{t,\mathrm{V}}), U_k(\mathbf{x}^0_{t,\mathrm{V}}))$ and observe y_t.
end for

The interval widths determine how long to explore.

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end for

The current verified features determine the buyer-specific interval. The current report does not enter the exploration price.

Estimation and exploitation

Algorithm 1: continue episode k

1: Compute $\widehat{\boldsymbol{\theta}}_k$ from I_k by (10).
2: Compute $\widehat{g}_k, \widehat{g}'_k$ from I_k by (12).
3: Construct the endpoint functions L_{k+1}, U_{k+1} using $\widehat{\boldsymbol{\theta}}_k$ by (6).
4: **for** $t \in I'_k$ **do**
5: Observe $\mathbf{x}_t$ and post

$$p_t = \min\Big\{B, \max\Big\{0, \widehat{g}_k\Big(\widehat{\boldsymbol{\theta}}_k^\top \mathbf{x}_t + \widehat{\boldsymbol{\beta}}_{\mathrm{R},k}^\top \mathbf{A}^{-1} \widehat{\boldsymbol{\beta}}_{\mathrm{R},k} \widehat{g}'_k(\widehat{\boldsymbol{\theta}}_k^\top \mathbf{x}_t)\Big)\Big\}\Big\}. \tag{4}$$

6: Observe y_t.
7: **end for**

If a required pricing-map evaluation fails, post $B/2$.

The same exploration observations support both learning steps.

◂ Exploration

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If a required pricing-map evaluation fails, post $B/2$.

The strategic correction uses the estimated valuation and pricing slope.

◂ Exploration

Using $\widehat{\boldsymbol{\theta}}_k$, SALE constructs the next episode's endpoint functions from:

- The estimated valuation from verified features $\mathbf{x}_{\mathrm{V}}$.
- Every possible true reportable profile $\mathbf{q} \in \mathcal{B}_{\mathrm{R}}$.

$$\varepsilon_k^{\mathrm{loc}} = \underbrace{4C_\theta W_x s_0 \lambda_k}_{\text{parameter uncertainty}} + \underbrace{\delta_z}_{\text{market noise}} . \tag{5}$$

$$L_{k+1}(\mathbf{x}_{\mathrm{V}}) = \max\left\{0, \widehat{\beta}_{\mathrm{V},k}^{\top}\mathbf{x}_{\mathrm{V}} + \widehat{\alpha}_k + \min_{\mathbf{q}\in\mathcal{B}_{\mathrm{R}}} \widehat{\beta}_{\mathrm{R},k}^{\top}\mathbf{q} - \varepsilon_k^{\mathrm{loc}}\right\},$$
$$U_{k+1}(\mathbf{x}_{\mathrm{V}}) = \min\left\{B, \widehat{\beta}_{\mathrm{V},k}^{\top}\mathbf{x}_{\mathrm{V}} + \widehat{\alpha}_k + \max_{\mathbf{q}\in\mathcal{B}_{\mathrm{R}}} \widehat{\beta}_{\mathrm{R},k}^{\top}\mathbf{q} + \varepsilon_k^{\mathrm{loc}}\right\}. \tag{6}$$

Verified features personalize the interval. The current report is not used to construct it.

At time $t \in I_{k+1}$, evaluate at $\mathbf{x}_{t,\mathrm{V}}^0$ and draw uniformly. Reset to $[0, B]$ if $L_{k+1} > U_{k+1}$. W_x bounds feature coordinates.

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Learning reduces parameter uncertainty. When the ℓ_1 error bound holds, the interval covers all feasible valuations.

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◂ Algorithm 1

Width-Adaptive Exploration Duration

A narrower interval reduces the regret cost of each exploration price. SALE can therefore explore longer to improve estimation.

Before episode k, the endpoint functions are known but future verified features are not. SALE uses their largest normalized width:

$$\omega_k = \max\left\{\omega_{\min}, \frac{1}{B}\sup_{\mathbf{x}_V \in \mathcal{B}_V}\{U_k(\mathbf{x}_V) - L_k(\mathbf{x}_V)\}\right\}, \qquad \omega_{\min} = \min\{1, 2\delta_z/B\}. \tag{7}$$

Thus $B\omega_k$ bounds every buyer-specific width, including any reset to $[0, B]$. With $\Lambda_k = 1 + \log \ell_k + s_0 \log(d+1)$, set

$$a_k = \min\left\{\ell_k, \left\lceil c_a(\ell_k \Lambda_k)^{\frac{2m+3}{4m+1}} \omega_k^{-\frac{2m+8}{4m+1}} \right\rceil\right\}. \tag{8}$$

Smaller ω_k gives weakly longer exploration, subject to $a_k \le \ell_k$.

$m \ge 3$ is the noise smoothness order; $c_a > 0$ is fixed.

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Write $W_k(\mathbf{x}_{\mathrm{V}}) = U_k(\mathbf{x}_{\mathrm{V}}) - L_k(\mathbf{x}_{\mathrm{V}})$. The seller observes only $y_t \in \{0, 1\}$. Uniform exploration converts this feedback into an unbiased valuation signal:

$$\begin{aligned} Y_t &= L_k(\mathbf{x}^0_{t,\mathrm{V}}) + W_k(\mathbf{x}^0_{t,\mathrm{V}}) y_t, \\ \mathbb{E}[Y_t \mid \mathbf{x}^0_t, \mathcal{F}_{t-1}] &= \boldsymbol{\theta}_0^\top \mathbf{x}^0_t. \end{aligned} \tag{9}$$

The identity holds when the interval covers all feasible valuations, even under non-i.i.d. feature arrivals. SALE then estimates the sparse parameter by regularized least squares:

$$\widehat{\boldsymbol{\theta}}_k = \operatorname*{arg\,min}_{\boldsymbol{\theta} \in \Theta} \left\{ \frac{1}{a_k} \sum_{t \in I_k} (Y_t - \boldsymbol{\theta}^\top \mathbf{x}^0_t)^2 + \lambda_k \|\boldsymbol{\theta}\|_1 \right\}, \quad \Theta = \{\boldsymbol{\theta} : \|\boldsymbol{\theta}\|_1 \le W_\theta\}. \tag{10}$$

Here $\lambda_k = c_\lambda \sqrt{\{1 + \log \ell_k + \log(d+1)\}/a_k}$.

The same estimate supports current pricing and the next episode's exploration.

$\mathcal{F}_{t-1}$ is the seller's history through time $t-1$.

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After estimating valuations, SALE learns the unknown noise distribution from the same exploration observations. For $w_t(\boldsymbol{\theta}) = p_t - \boldsymbol{\theta}^\top \mathbf{x}_t^0$, the true residual satisfies $\mathbb{E}[y_t \mid w_t(\boldsymbol{\theta}_0) = u] = 1 - F(u)$.

$$\widehat{F}_k(u) = 1 - \frac{\sum_{t \in I_k} K((w_t(\widehat{\boldsymbol{\theta}}_k) - u)/b_k) y_t}{\sum_{t \in I_k} K((w_t(\widehat{\boldsymbol{\theta}}_k) - u)/b_k)}, \qquad b_k = (\omega_k / a_k)^{1/(2m+3)}. \tag{11}$$

Substituting $\widehat{F}_k$ into the pricing equation gives the price and its slope:

$$\widehat{\phi}_k(u) = u - \frac{1 - \widehat{F}_k(u)}{\widehat{F}_k^{(1)}(u)}, \qquad \widehat{g}_k(u) = u + \widehat{\phi}_k^{-1}(-u), \qquad \widehat{g}_k'(u) = \frac{d}{du}\widehat{g}_k(u). \tag{12}$$

- $\widehat{g}_k$ determines the price; $\widehat{g}_k'$ enters the strategic correction in (4).

K is an order-m kernel. Estimating g' requires F''. Inversion is safeguarded on $I_g = [\delta_z, B - \delta_z]$; a failed required evaluation gives price $B/2$.

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◂ Algorithm 1

SALE-U retains the exploration design and estimators of SALE-K. It learns the strategic adjustment from matched repeat-buyer reports:

$$\mathbf{x}_{i,\mathrm{R}} - \mathbf{x}^0_{i,\mathrm{R}} = \boldsymbol{\gamma}_0 g'(\boldsymbol{\theta}_0^\top \mathbf{x}_i), \qquad \boldsymbol{\gamma}_0 = -\mathbf{A}^{-1}\boldsymbol{\beta}_{\mathrm{R},0}. \tag{13}$$

The archive $\mathcal{M}$ stores one true–reported pair per buyer. The two visits may occur in either order.

$$\widehat{\boldsymbol{\gamma}}_t = \frac{\sum_{i\in\mathcal{M}} \widehat{g}'_i(\mathbf{x}_{i,\mathrm{R}} - \mathbf{x}^0_{i,\mathrm{R}})}{\sum_{i\in\mathcal{M}} (\widehat{g}'_i)^2}, \qquad \widehat{g}'_i = \widehat{g}'_k(\widehat{\boldsymbol{\theta}}_k^\top \mathbf{x}_i). \tag{14}$$

SALE-U uses this estimate when the slope estimates exist and the denominator is positive. Matched buyers can be priced using their stored true features.

The policy estimates $\boldsymbol{\gamma}_0$, not the full matrix $\mathbf{A}$.

Algorithm 2: full pseudocode

Algorithm 2 SALE-U (Unknown $\mathbf{A}$): exploration

Input: B and ℓ_0. Initialize $L_1 \equiv 0$, $U_1 \equiv B$, and $\mathcal{E}_1, \mathcal{E}_2, \mathcal{M} \leftarrow \emptyset$.
for each episode $k = 1, 2, \ldots$ **do**
  Set ℓ_k, S_k, I_k, I'_k as in Algorithm 1, with exploration duration a_k from (8).
  for $t \in I_k$ **do**
    Observe $\mathbf{x}_t^0$ and id_t; draw $p_t \sim \mathrm{Unif}(L_k(\mathbf{x}_{t,\mathrm{V}}^0), U_k(\mathbf{x}_{t,\mathrm{V}}^0))$ and observe y_t.
    If $\mathrm{id}_t \notin \mathcal{E}_1$, store $(\mathrm{id}_t, \mathbf{x}_t^0)$ in $\mathcal{E}_1$.
    If $\mathrm{id}_t \in \mathcal{E}_2$ and $\mathrm{id}_t \notin \mathcal{M}$, add the true–reported pair to $\mathcal{M}$.
  end for
  Compute $\widehat{\boldsymbol{\theta}}_k$ by (10) and $\widehat{g}_k, \widehat{g}'_k$ by (12).
  Construct L_{k+1}, U_{k+1} by (6).
  Continue with exploitation on the next slide.
end for

$\mathcal{E}_1, \mathcal{E}_2$ store first exploration and exploitation records. Visits with the same ID share the same true features.

Exploitation and pricing
◂ SALE-U overview

Algorithm 2: exploitation (continued)

For each $t \in I'_k$**:**
Observe $\mathbf{x}_t$ and id_t.
If $\mathrm{id}_t \notin \mathcal{E}_2$, store $(\mathrm{id}_t, \mathbf{x}_t)$ in $\mathcal{E}_2$.
If $\mathrm{id}_t \in \mathcal{E}_1$ and $\mathrm{id}_t \notin \mathcal{M}$, add the true–reported pair to $\mathcal{M}$.
if $\mathrm{id}_t \in \mathcal{M}$ **then**
    Retrieve $\mathbf{x}_t^0$ from $\mathcal{E}_1$ and post $p_t = \min\{B, \max\{0, \widehat{g}_k(\widehat{\boldsymbol{\theta}}_k^\top \mathbf{x}_t^0)\}\}$.
else if the estimator in (14) is available **then**
    Compute $\widehat{\gamma}_t$ by (14) and post

$$p_t = \min\left\{B, \max\left\{0, \widehat{g}_k\left(\widehat{\boldsymbol{\theta}}_k^\top \mathbf{x}_t - \widehat{\boldsymbol{\beta}}_{\mathrm{R},k}^\top \widehat{\gamma}_t \, \widehat{g}'_k(\widehat{\boldsymbol{\theta}}_k^\top \mathbf{x}_t)\right)\right\}\right\}. \tag{15}$$

else
    Post $p_t = \min\{B, \max\{0, \widehat{g}_k(\widehat{\boldsymbol{\theta}}_k^\top \mathbf{x}_t)\}\}$.
end if
Observe y_t.

If a required pricing-map evaluation fails, post $B/2$.

◂ Algorithm 2 ◂ SALE-U overview

Sparse valuation learning

The population equation converts binary feedback into an unbiased signal. Martingale concentration supports sparse estimation under non-i.i.d. arrivals.

Nonparametric pricing-map learning

The noise estimator uses estimated valuations from the same observations. Localized empirical-process analysis controls this error in high dimension.

Confidence recovery

An estimation error may invalidate a later exploration interval. The recovery argument bounds the expected cost of these episodes.

Together, these results yield an unconditional expected regret bound.

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Together, these results yield an unconditional expected regret bound.

Assumption 1 (Bounded covariates and sparse valuation)

The seller knows rectangular sets $\mathcal{B}_{\mathrm{V}}, \mathcal{B}_{\mathrm{R}}$ containing the true features, with coordinate bound $W_x \geq 1$. Moreover,

$$\|\mathbf{x}_t^0\|_\infty \leq W_x, \qquad \boldsymbol{\theta}_0 \in \Theta, \qquad 1 \leq s_0 = \|\boldsymbol{\theta}_0\|_0 \leq d+1.$$

Assumption 2 (Predictable context stability)

Arrivals are independent of valuation noises and policy random draws. For $\Sigma_t = \mathbb{E}[\mathbf{x}_t^0 \mathbf{x}_t^{0\top} \mid \mathcal{F}_{t-1}]$, almost surely,

$$\lambda_{\min} I \preceq \Sigma_t \preceq \lambda_{\max} I, \qquad \|\Sigma_t - \Sigma_s\|_{\mathrm{op}} \leq \eta_\Sigma \lambda_{\min} \quad (s,t \geq 1), \tag{16}$$

where $\eta_\Sigma < \rho_z^{-2} - 1$ and $\rho_z = \max\{1/2, 1 - \delta_z/(2W_x W_\theta)\}$.

The eigenvalue bounds support estimation; the drift bound enters recovery.

Assumption 3 (Noise smoothness)

For an integer $m \geq 3$, F is m times continuously differentiable on the interior of its support. Its mth derivative is Lipschitz, and F' is bounded away from zero on every compact subinterval.

Assumption 4 (Regular virtual valuation and interior oracle residual)

For some $c_\phi > 0$,

$$\phi'(z) \geq c_\phi \quad (z \in (-\delta_z, \delta_z)), \qquad g(u) - u \in (-\delta_z, \delta_z) \quad (u \in I_g),$$

where $\phi(z) = z - \{1 - F(z)\}/F'(z)$ and $I_g = [\delta_z, B - \delta_z]$.

Noise smoothness supports estimation of F, F', F''. Regularity of the pricing equation permits stable inversion.

Assumption 5 (Manipulation cost and interior response)

The matrix $\mathbf{A}$ is symmetric positive definite. For every optimal report, $\boldsymbol{\theta}_0^\top \mathbf{x}_t$ lies in a fixed compact subset of $\operatorname{int}(I_g)$.

Assumption 6 (Kernel)

The kernel satisfies

$$\int K(s)\,ds = 1, \qquad \int s^j K(s)\,ds = 0 \quad (j = 1, \ldots, m-1), \qquad \int |s^m K(s)|\,ds < \infty.$$

Moreover, K, K', K'' are Lipschitz and supported on $[-1, 1]$ after a fixed rescaling if necessary.

The response condition keeps strategic correction in the regular pricing region. The kernel condition concerns the estimator selected by the seller.

For Lemma 1, choose the penalty and exploration length so that

$$\begin{aligned} \lambda_k &\geq B_{\mathrm{sp}} W_x (B + W_x W_\theta) \sqrt{\frac{1 + \log \ell_k + \log(d+1)}{a_k}}, \\ a_k &\geq \frac{B_{\mathrm{sp}} W_x^4 s_0^2}{\lambda_{\min}^2} \{1 + \log \ell_k + \log(d+1)\}. \end{aligned} \tag{17}$$

For Lemmas 2 and 3, respectively,

$$a_k b_k^5 \geq B_{\mathrm{np}} \Lambda_k, \quad a_k b_k^2 \geq B_{\mathrm{np}} s_0^2 \{1 + \log \ell_k + \log(d+1)\}, \tag{18}$$

$$a_k b_k^5 \geq B_{\mathrm{map}} \Lambda_k, \quad a_k b_k^2 \geq B_{\mathrm{map}} s_0^2 \{1 + \log \ell_k + \log(d+1)\}. \tag{19}$$

These thresholds depend on fixed regularity, kernel, and tuning constants. Taking $C_\theta \geq 3/\lambda_{\min}$ covers the sparse error bounds.

Let $\mathcal{H}_{k-1} = \mathcal{F}_{S_k-1}$. The event $\mathcal{G}_k$ means that the exploration intervals cover every valuation allowed by the model.

Lemma 1 (Sparse valuation estimation)

Under Assumptions 1–2 and the penalty and sample-size conditions in (17), on $\mathcal{G}_k$, conditionally on $\mathcal{H}_{k-1}$, with probability at least $1 - 2\ell_k^{-2}$,

$$\|\widehat{\boldsymbol{\theta}}_k - \boldsymbol{\theta}_0\|_2 \le \frac{3\lambda_k\sqrt{s_0}}{\lambda_{\min}}, \qquad \|\widehat{\boldsymbol{\theta}}_k - \boldsymbol{\theta}_0\|_1 \le \frac{12 s_0 \lambda_k}{\lambda_{\min}}. \tag{20}$$

The population equation (9) and martingale concentration support estimation under non-i.i.d. arrivals.

- The ℓ_2 bound controls the error passed to noise estimation.
- The ℓ_1 bound certifies the next exploration interval.

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The population equation (9) and martingale concentration support estimation under non-i.i.d. arrivals.

- The ℓ_2 bound controls the error passed to noise estimation.
- The ℓ_1 bound certifies the next exploration interval.

Localized exploration generates residual-design density $q_k = a_k^{-1} \sum_{t \in I_k} W_k(\mathbf{x}_{t,\mathrm{V}}^0)^{-1} \geq 1/(B\omega_k)$. Let J be a fixed compact interval in $(-\delta_z, \delta_z)$ whose interior contains all oracle residuals $g(u) - u$, $u \in I_g$.

Lemma 2 (Source-aware market-noise estimation)

Under the assumptions and penalty condition of Lemma 1, Assumptions 3 and 6, and (18), on $\mathcal{G}_k$, conditionally on $\mathcal{H}_{k-1}$, with probability at least $1 - 3\ell_k^{-2}$,

$$\|\widehat{F}_k - F\|_{\infty,J} \leq C_{\mathrm{np}} \left\{ \frac{\lambda_k \sqrt{s_0}}{\lambda_{\min}} + b_k^m + \sqrt{\frac{\Lambda_k}{a_k b_k q_k}} \right\}, \tag{21}$$

$$\|\widehat{F}_k^{(1)} - F'\|_{\infty,J} \leq C_{\mathrm{np}} \left\{ \frac{\lambda_k \sqrt{s_0}}{\lambda_{\min}} + b_k^{m-1} + \sqrt{\frac{\Lambda_k}{a_k b_k^3 q_k}} \right\}, \tag{22}$$

$$\|\widehat{F}_k^{(2)} - F''\|_{\infty,J} \leq C_{\mathrm{np}} \left\{ \frac{\lambda_k \sqrt{s_0}}{\lambda_{\min}} + b_k^{m-1} + \sqrt{\frac{\Lambda_k}{a_k b_k^5 q_k}} \right\}. \tag{23}$$

Uniform control is established before substituting the random sparse estimate. The derivative bounds have the same bias order but different stochastic terms.

Lemma 3 (Source-aware pricing-map estimation)

Under Lemma 2, Assumption 4, and (19), on $\mathcal{G}_k$, conditionally on $\mathcal{H}_{k-1}$, with probability at least $1 - 3\ell_k^{-2}$,

$$\|\widehat{g}_k - g\|_{\infty, I_g} \le C_{\mathrm{map}} \left\{ \frac{\lambda_k \sqrt{s_0}}{\lambda_{\min}} + b_k^{m-1} + \sqrt{\frac{\Lambda_k}{a_k b_k^3 q_k}} \right\}, \tag{24}$$

$$\|\widehat{g}'_k - g'\|_{\infty, I_g} \le C_{\mathrm{map}} \left\{ \frac{\lambda_k \sqrt{s_0}}{\lambda_{\min}} + b_k^{m-1} + \sqrt{\frac{\Lambda_k}{a_k b_k^5 q_k}} \right\}. \tag{25}$$

Sparse estimation error + kernel bias + stochastic error.

The estimator reuses the observations that produced $\widehat{\boldsymbol{\theta}}_k$. Localized empirical-process analysis controls the resulting nonparametric error.

Lemma 3 (Source-aware pricing-map estimation)

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Sparse estimation error + kernel bias + stochastic error.

Localization improves the stochastic error bound: the normalized width ω_k gives the density bound $q_k \ge 1/(B\omega_k)$.

Lemma 4: Recovery of Confidence Validity

Set $\overline{\Sigma}_k = a_k^{-1} \sum_{t \in I_k} \Sigma_t$ and $e_k = \|\widehat{\boldsymbol{\theta}}_k - \boldsymbol{\theta}_0\|_{\overline{\Sigma}_k}$.

Lemma 4 (Recovery of confidence validity)

Under Assumptions 1–2, $C_\theta \geq 3/\lambda_{\min}$, and sufficiently large c_λ, there are $\rho_{\mathrm{g}}, \rho_{\mathrm{l}} \in (0,1)$ such that, for each $k \geq 2$, conditionally on $\mathcal{H}_{k-1}$, jointly with probability at least $1 - 2\ell_k^{-2}$,

$$e_k \leq \rho_{\mathrm{g}} e_{k-1} + C_{\mathrm{dyn}} \sqrt{\lambda_k}, \tag{26}$$

$$e_k \leq \rho_{\mathrm{l}} e_{k-1} + C_{\mathrm{dyn}} \sqrt{s_0} \lambda_k \quad \text{if } e_{k-1} \leq c_{\mathrm{nbhd}} (d+1)^{-1/2}. \tag{27}$$

If $e_{k-1} \leq c_{\mathrm{nbhd}} \min\{(d+1)^{-1/2}, \sqrt{s_0}\lambda_{k-1}\}$, then $\boldsymbol{\theta}_0 \in \mathcal{C}_k$. Under exploration over $[0, B]$, the first term in (26) is absent.

Here $\mathcal{C}_k = \{\boldsymbol{\theta} \in \Theta : \|\boldsymbol{\theta} - \widehat{\boldsymbol{\theta}}_k\|_2 \leq C_\theta \lambda_k \sqrt{s_0},\ \|\boldsymbol{\theta} - \widehat{\boldsymbol{\theta}}_k\|_1 \leq 4 C_\theta s_0 \lambda_k\}$. Constants may depend on the fixed noise distribution through its endpoint tails.

The preceding estimation bounds require valid exploration intervals. An inaccurate estimate can invalidate the next interval, so high-probability estimation alone is not enough.

Lemma 5 (Finite expected recovery cost)

Under Lemmas 4 and 3, suppose the penalty and sample-size conditions of Lemmas 1–3 hold for all $k \geq k_\star$, where $k_\star \geq 2$ is deterministic. Then

$$\mathbb{E}\left[\sum_{k \geq k_\star} \ell_k \mathbb{I}\{\mathcal{J}_k^c\}\right] \leq C_{\mathrm{rec}}. \tag{28}$$

Here $\mathcal{J}_k$ is the event that the preceding confidence set covers $\boldsymbol{\theta}_0$ and the estimation bounds of Lemmas 1–3 hold in episode k.

C_{rec} does not depend on $k_\star, T, d$, or s_0.

Lemma 4 establishes contraction back to confidence validity. The finite expected recovery cost enters the unconditional regret bound.

Recovering from a Missed Interval

The preceding estimation bounds require valid exploration intervals. An inaccurate estimate can invalidate the next interval, so high-probability estimation alone is not enough.

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Repeat Buyers and Usable Matched Records

For exploitation in episode k, let $N_k = \sum_{j \le k} a_j$ and let n_t count distinct matched buyers. The effective rate is the largest $\tau \in [0, 1]$ satisfying

$$\max\{1, n_t\} \ge \tau N_k \qquad \text{almost surely for all relevant } k, t.$$

The expected-regret analysis also admits the manuscript's conditional inverse-count condition:

$$\mathbb{E}\left[\frac{N_k}{\max\{1, n_t\}} \,\middle|\, \mathcal{F}_{S_k + a_k - 1}\right] \le \frac{1}{\tau}.$$

The numerical return frequencies $\tau_{\text{sim}} \in \{0.05\%, 0.1\%\}$ determine two-visit paths. They need not equal this effective rate.

A pair is usable only when the visits occur in different phases; both arrival orders are supported.

Lemma 6: Learning the Strategic Adjustment

Lemma 6 requires a threshold $B_\gamma \geq B_{\mathrm{map}}$ such that

$$\begin{aligned} a_k b_k^5 &\geq B_\gamma \Lambda_k, \\ a_k b_k^2 &\geq B_\gamma s_0^2 \{1 + \log \ell_k + \log(d+1)\}. \end{aligned} \tag{29}$$

B_γ may depend on the fixed model, tuning, and margin constants, including the report bound induced by $\mathbf{A}$ and the derivative margin, but not on k, T, d, s_0, or τ.

Write $\rho_A = \max\{1, \|\mathbf{A}^{-1}\|_{\mathrm{op}}\}$.

Lemma 6 (Estimation of γ_0)

Under Lemma 3, Assumption 5, $\tau > 0$, $|g'(u)| \geq c_g > 0$ over matched records, and (29), the probability that $\mathcal{G}_k$ occurs and the following bound fails at any eligible $t \in I'_k$ is at most $4\ell_k^{-2}$:

$$\|\widehat{\boldsymbol{\gamma}}_t - \boldsymbol{\gamma}_0\|_2^2 \leq \frac{C_\gamma \rho_A^4}{\tau} \left\{ \frac{s_0 \lambda_k^2}{\lambda_{\min}^2} + b_k^{2m-2} + \frac{\Lambda_k}{a_k b_k^5 q_k} \right\}. \tag{30}$$

A time is eligible when the matched-pair estimator is available.

C_γ does not absorb ρ_A or τ. The factor $1/\tau$ records the cost of using the smaller matched sample.

Write $\alpha_m = (2m+3)/(4m+1)$, $\kappa_m = (6m-6)/(4m+1)$, and $\Lambda_T = 1 + \log T + s_0 \log(d+1)$. Let K_T be the last episode beginning by time T. For $\pi \in \{\mathrm{K}, \mathrm{U}\}$, define

$$\omega_{\pi,T} = \left[\frac{\mathbb{E}_\pi \sum_{k=1}^{K_T} (\ell_k \Lambda_k)^{\alpha_m} \omega_k^{\kappa_m}}{\sum_{k=1}^{K_T} (\ell_k \Lambda_k)^{\alpha_m}} \right]^{1/\kappa_m}. \tag{31}$$

- This is the regret-weighted power mean of the episode widths.
- $\omega_{\min} \le \omega_{\pi,T} \le 1$; global exploration gives $\omega_{\pi,T} = 1$.
- It is small when intervals are narrow in episodes carrying substantial regret weight.

This quantity summarizes the gain in Theorems 1–2; it is not a policy input.

The leading regret bounds apply when

$$
\begin{aligned}
(T\Lambda_T)^{\frac{2m+3}{4m+1}} &\geq c_0[1+s_0\log(d+1)]^{\frac{2m+3}{2m-2}}, \\
(T\Lambda_T)^{\frac{2m+3}{4m+1}} &\geq c_0(1+s_0)^{\frac{6m+1}{2m+1}}[1+\log(d+1)]^{\frac{2m}{2m+1}}.
\end{aligned}
\tag{32}
$$

- These conditions absorb the initial estimation and recovery costs into the leading regret term.
- They are conditions for the bound, not policy inputs.
- For SALE-U, c_0 may be enlarged without depending on τ.

Theorem 1 (Regret of SALE-K)

Under Assumptions 1–6, let $C_\theta \geq 3/\lambda_{\min}$ and choose c_λ sufficiently large. For every T satisfying (32), SALE-K has expected regret

$$\mathrm{Regret}_{\mathrm{K}}(T) \leq C_{\mathrm{K}} \rho_A^4 \omega_{\mathrm{K},T}^{\kappa_m} (T\Lambda_T)^{\alpha_m}. \tag{33}$$

Here $\omega_{\mathrm{K},T}$ is the effective acquisition width in (31).

- $\alpha_m < 1$ gives sublinear dependence on T; dimension enters through $\log(d+1)$ in Λ_T.
- $\omega_{\mathrm{K},T}^{\kappa_m}$ measures the improvement from source-aware exploration.
- The bound includes recovery costs and does not condition on permanent coverage.

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- $\omega_{\mathrm{K},T}^{\kappa_m}$ measures the improvement from source-aware exploration.
- The bound includes recovery costs and does not condition on permanent coverage.

The effective repeat-buyer rate τ measures the availability of matched records; it is not a policy input.

Theorem 2 (Regret of SALE-U)

Under the assumptions and tuning of Theorem 1, suppose $\tau > 0$ and $|g'(u)| \geq c_g > 0$ over matched records. For every T satisfying (32), after a fixed enlargement of c_0 if needed,

$$\mathrm{Regret}_{\mathrm{U}}(T) \leq \rho_A^4 \left(C_{\mathrm{U},0} + \frac{C_{\mathrm{U},1}}{\tau} \right) \omega_{\mathrm{U},T}^{\kappa_m} (T\Lambda_T)^{\alpha_m}. \tag{34}$$

The constants $C_{\mathrm{U},0}$ and $C_{\mathrm{U},1}$ do not absorb ρ_A or τ.

Learning the strategic adjustment preserves the order in T, d, s_0, m. The additional factor $1/\tau$ reflects the smaller matched sample.

The Value of Source-Aware Exploration

Blind ETC uses the same estimators and exploitation rule, but draws exploration prices from $[0, B]$ and sets $\omega_k = 1$.

SALE replaces the unit width factor in the corresponding leading bound by

$$\omega_{\pi,T}^{\kappa_m}, \qquad \kappa_m = \frac{6m-6}{4m+1} > 0.$$

- A narrower interval reduces per-period exploration loss.
- Width adaptation permits a longer exploration phase.
- After balancing these effects, the regret bound improves by a positive power of the effective width.

The gain depends on how much verified features reduce uncertainty about valuation.

The comparison concerns regret upper bounds, not pathwise dominance. For SALE-U and Blind ETC-U, it uses a common positive lower bound on their effective repeat-buyer rates.

Experimental Setup

We evaluate SALE through controlled simulations and a semi-synthetic AutoLoan study.

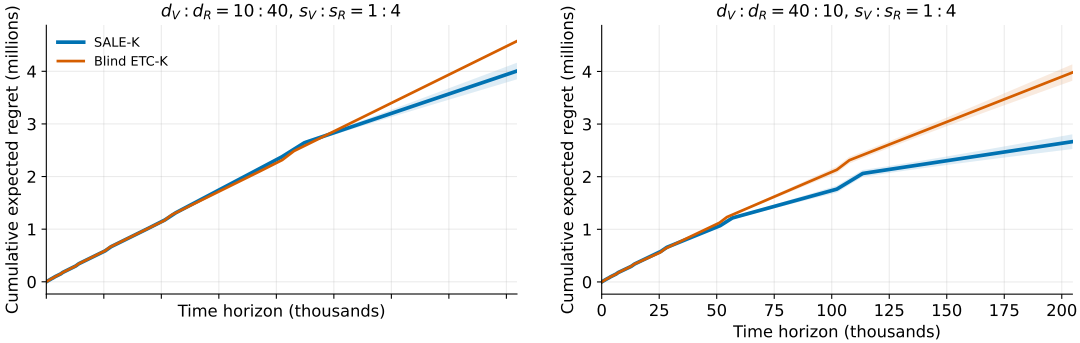
Comparison	**Purpose**
SALE and Blind ETC	Isolate localized exploration and its width-adaptive schedule
SALE-K and RMLP-2	Compare with a high-dimensional parametric reference applied to reported features
Semiparametric and Gaussian pricing maps	Isolate the contribution of learning the pricing function
SALE-U and Blind ETC-U	Examine localization when the strategic adjustment is estimated from repeat buyers

Curves report cumulative expected regret over the full horizon. Policies within a panel use common feature, noise, and price-randomization streams.

Shaded regions show one Monte Carlo standard error.

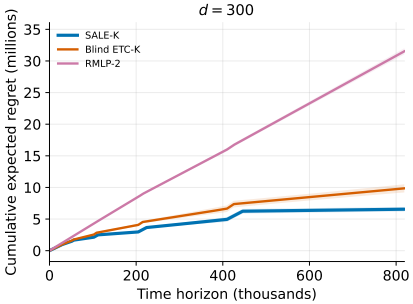
The Value of Source-Aware Exploration

Fix $d = 50$, $s_0 = 6$ and active slopes $(s_{\mathrm{V}}, s_{\mathrm{R}}) = (1, 4)$. Vary the numbers of verified and reportable coordinates.



Localization becomes pronounced when most candidate coordinates are verified.

$T = 204{,}600$, $B = 100$; 50 replications. Total signal energy, verified signal share, and manipulation strength remain fixed.



High-dimensional strategic pricing

SALE-K and Blind ETC-K use the same estimators and exploitation rule.

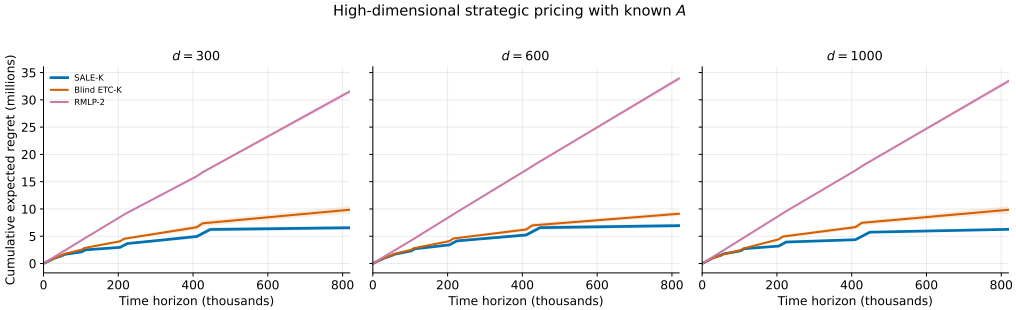
RMLP-2 uses a standard-Gaussian pricing map applied to reported features.

$d \in \{300, 600, 1000\}$, $s_0 = 6$; the first $4d/5$ coordinates are verified.

SALE-K has the lower terminal regret in all three designs.

$B = 140$, $T = 819{,}000$; 20 replications. RMLP-2 uses a standard-Gaussian pricing map.

Known Manipulation Cost

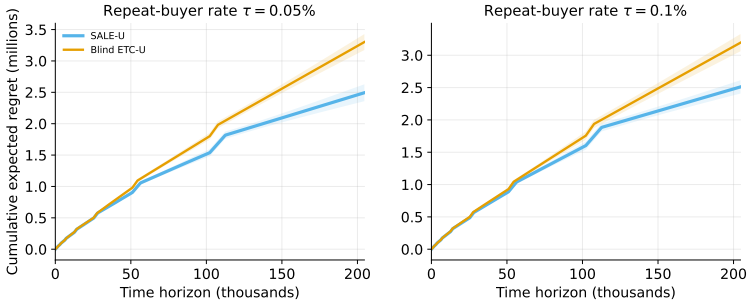


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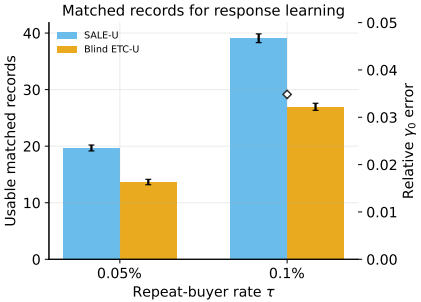
Unknown Manipulation Cost



Only returns across exploration and exploitation form usable matched records.

SALE-U has lower terminal regret at both two-visit rates.

$d = 50$, $s_0 = 6$, $B = 100$, $T = 204{,}600$; 50 replications. The percentages refer to the exogenous two-visit rate τ_{sim}, distinct from the effective rate τ.



Learning the strategic adjustment

The nominal repeat-buyer rate determines return frequency. Only cross-phase visits supply usable matched observations.

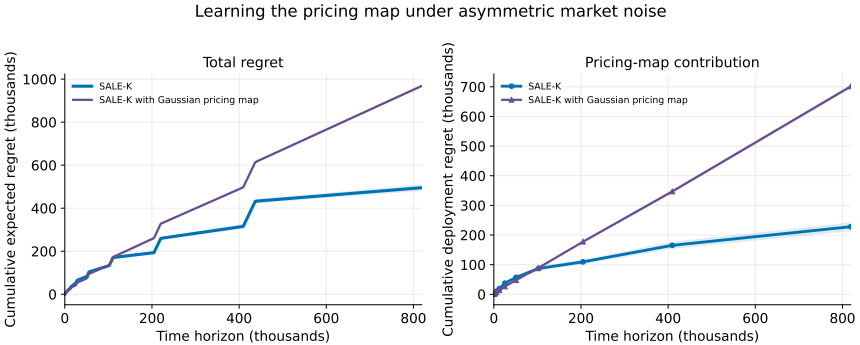
The plot reports matched records and estimation error separately.

Only returns across exploration and exploitation form usable matched records.

SALE-U has lower terminal regret at both two-visit rates.

$d = 50$, $s_0 = 6$, $B = 100$, $T = 204{,}600$; 50 replications. The percentages refer to the exogenous two-visit rate τ_{sim}, distinct from the effective rate τ.

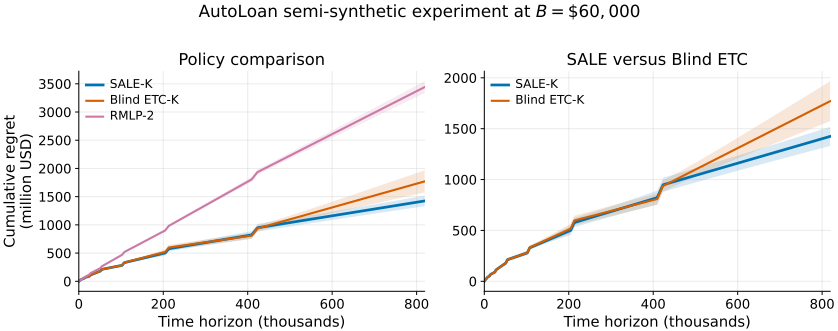
Unknown Market Noise



The comparator replaces only SALE-K's semiparametric pricing map by a standard-Gaussian working model.

The policies separate after longer exploitation phases.

Smooth, bounded, mean-zero asymmetric noise; $d = 2$, $m = 3$, $B = 15$, $T = 819{,}000$; 50 replications.



$d_{\mathrm{V}} = 64$, $d_{\mathrm{R}} = 2$, $s_0 = 8$; online contexts are sampled from the held-out empirical distribution.

SALE-K ends below Blind ETC-K as personalized exploration becomes less costly.

$B = \$60{,}000$, $T = 819{,}000$; 30 replications. The results concern a held-out, semi-synthetic valuation model and are not field evidence.

Conclusion

We study contextual dynamic pricing when some buyer features are verified, others may be strategically reported, and the market-noise distribution is unknown.

SALE

- Uses verified features to localize randomized exploration prices.
- Learns a sparse valuation model and a semiparametric pricing map.
- Corrects for strategic manipulation during exploitation.

Theory and numerical evidence

- Sublinear expected regret with an explicit multiplicative improvement when personalized intervals are narrow.
- Matched repeat-buyer records preserve the regret order when the manipulation cost is unknown.
- Controlled simulations and a semi-synthetic AutoLoan study examine these mechanisms.

Thank you